DL and Tensor Mathematics

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1 Introduction

Abstract

We present a complete mathematical formulation for **Tensor-Train Dictionary Learning (TT-DL)**, a framework that replaces the exponentially costly Tucker core tensor with a linearly scaling Tensor-Train structure for sparse representation of high-order data. The document covers: (i) the TT-DL optimization problem; (ii) a DMRG-inspired alternating minimization algorithm with detailed derivations of the sparse coding and dictionary update sub-problems; (iii) an adaptive rank selection mechanism based on SVD truncation; and (iv) a deep unrolling extension that converts the iterative algorithm into a trainable neural network. We provide a thorough complexity analysis showing that TT-DL achieves *linear* parameter scaling in the tensor order N, compared to the *exponential* scaling of Tucker-based approaches.

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2 Introduction and Motivation

Dictionary learning (DL) seeks a redundant set of atoms—the *dictionary*—under which observed signals admit sparse representations. For multidimensional (tensor) data, *tensor dictionary learning* extends this idea by learning separable mode dictionaries, one per tensor mode, together with sparse coefficient structures. The dominant paradigm uses Tucker decomposition to couple the mode dictionaries through a core tensor.

The fundamental limitation. The Tucker core tensor has $\prod_{n=1}^{N} J_n$ entries, where J_n is the dictionary size along mode n. For $N \geq 5$ this is computationally intractable:

Tensor Order N	**Tucker Core Size** (J=15)	**TT Parameters** (J=15, r=5)
4	50,625	1,500
6	11,390,625	2,250
8	2.56×10^9	3,000
10	5.77×10^{11}	3,750

This document develops **Tensor-Train Dictionary Learning (TT-DL)**, which replaces the Tucker core with a Tensor-Train (TT) structure whose parameter count scales *linearly* in N.

3 Background: Classical Dictionary Learning

Given data $\{y_1, \ldots, y_m\} \subset R^n$, find a dictionary $D \in R^{n \times K}$ ($K \geq n$) and sparse coefficients $S = [s_1 \mid \cdots \mid s_m] \in R^{K \times m}$ solving

$$\min_{D,\,S} \ \frac{1}{2}\big\|Y - DS\big\|_{\mathrm{F}}^2 + \lambda \|S\|_1 \qquad s.t. \quad \|d_k\|_2 \leq 1, \ \ k = 1, \ldots, K, \tag{1}$$

where $Y = [y_1 | \cdots | y_m]$ and $\lambda > 0$ controls sparsity.
The standard approach is **alternating minimization**:

1. **Sparse Coding.** Fix D; for each sample i solve the LASSO:

$$\min_{s_i} \ \frac{1}{2}\|y_i - Ds_i\|_2^2 + \lambda \|s_i\|_1. \tag{2}$$

2. **Dictionary Update.** Fix S; solve the constrained least-squares problem:

$$\min_{D} \ \frac{1}{2}\|Y - DS\|_{\mathrm{F}}^2 \qquad s.t. \quad \|d_k\|_2 \leq 1. \tag{3}$$

4 Tucker Dictionary Learning (Baseline)

For N-th order data tensors $\mathcal{Y}_i \in R^{I_1 \times \cdots \times I_N}$, Tucker DL learns mode dictionaries $D^{(n)} \in R^{I_n \times J_n}$ for $n = 1, \ldots, N$ and sparse core tensors $\mathcal{G}_i \in R^{J_1 \times \cdots \times J_N}$ such that

$$\mathcal{Y}_i \approx \mathcal{G}_i \times_1 D^{(1)} \times_2 D^{(2)} \cdots \times_N D^{(N)}. \tag{4}$$

Tucker DL Optimization:

$$\min_{D^{(1)},\ldots,D^{(N)}\{\mathcal{G}_i\}_{i=1}^m} \sum_{i=1}^{m} \frac{1}{2} \left\| \mathcal{Y}_i - \mathcal{G}_i \times_1 D^{(1)} \cdots \times_N D^{(N)} \right\|_{\mathrm{F}}^2 + \lambda \sum_{i=1}^{m} \|\mathcal{G}_i\|_1 \tag{5}$$

subject to $\quad \|d_k^{(n)}\|_2 \leq 1 \quad \forall\, n = 1, \ldots, N, \;\; k = 1, \ldots, J_n.$

Key Insight: The core tensor $\mathcal{G}_i$ has $\prod_{n=1}^{N} J_n$ entries—**exponential** in the tensor order N. This makes Tucker DL intractable for $N5$.

5 Tensor-Train Decomposition

Definition (Tensor-Train Format). A tensor $\mathcal{T} \in R^{I_1 \times \cdots \times I_N}$ is in TT format if every entry can be written as

$$\mathcal{T}(i_1, i_2, \ldots, i_N) = G^{(1)}(i_1)\, G^{(2)}(i_2)\, \cdots\, G^{(N)}(i_N), \tag{6}$$

where each $G^{(n)}(i_n)$ is an $r_{n-1} \times r_n$ matrix for fixed i_n, equivalently a 3rd-order core tensor $G^{(n)} \in R^{r_{n-1} \times I_n \times r_n}$, with boundary conditions $r_0 = r_N = 1$. The integers $r_1, \ldots, r_{N-1}$ are called **TT-ranks** (bond dimensions).

Storage cost.

$$Totalparameters = \sum_{n=1}^{N} r_{n-1}\, I_n\, r_n. \tag{7}$$

For uniform ranks r and mode sizes I, this is NIr^2—**linear in** N.

Mode-n unfoldings. Each TT-core $G^{(n)} \in R^{r_{n-1} \times I_n \times r_n}$ has two matrix unfoldings: Left unfolding: $\mathrm{G}_{(1)}^{(n)} \in R^{(r_{n-1} \cdot I_n) \times r_n}$,
$Rightunfolding:\;\; G_{(2)}^{(n)} \in R^{r_{n-1} \times (I_n \cdot r_n)}$.

6 TT-Dictionary Learning: Problem Formulation

6.1 Model

Each data tensor $\mathcal{Y}_i \in R^{I_1 \times \cdots \times I_N}$ is modelled as

$$\mathcal{Y}_i \approx \mathcal{S}_i \times_1 D^{(1)} \times_2 D^{(2)} \cdots \times_N D^{(N)}, \tag{8}$$

where the sparse coefficient tensor $\mathcal{S}_i \in R^{J_1 \times \cdots \times J_N}$ is in TT format:

$$\mathcal{S}_i(j_1, \ldots, j_N) = C_i^{(1)}(j_1)\, C_i^{(2)}(j_2)\, \cdots\, C_i^{(N)}(j_N), \tag{9}$$

with TT-cores $C_i^{(n)} \in R^{\rho_{n-1} \times J_n \times \rho_n}$ and TT-ranks $\rho = (\rho_1, \ldots, \rho_{N-1})$.

6.2 Dictionary absorption

Define absorbed cores by contracting each dictionary into its corresponding TT-core:

$$\hat{C}_i^{(n)}(\alpha_{n-1},\, i_n,\, \alpha_n) = \sum_{j_n=1}^{J_n} D^{(n)}(i_n,\, j_n)\, C_i^{(n)}(\alpha_{n-1},\, j_n,\, \alpha_n). \tag{10}$$

The reconstructed tensor is itself a TT:

$$\mathcal{Y}_i(i_1, \ldots, i_N) \approx \hat{C}_i^{(1)}(i_1)\, \hat{C}_i^{(2)}(i_2)\, \cdots\, \hat{C}_i^{(N)}(i_N). \tag{11}$$

6.3 Optimization problem

TT-DL Optimization (Project 1):

$$\min_{D^{(1)},\ldots,D^{(N)}\{C_i^{(n)}\}} \underbrace{\sum_{i=1}^{m} \frac{1}{2} \big\| \mathcal{Y}_i - \mathrm{Recon}\big(D^{(1:N)},\, C_i^{(1:N)}\big) \big\|_{\mathrm{F}}^2}_{data fidelity} + \underbrace{\lambda \sum_{i=1}^{m} \sum_{n=1}^{N} \big\| C_i^{(n)} \big\|_1}_{sparsity} \tag{12}$$

$$\text{subject to} \quad \|d_k^{(n)}\|_2 \le 1 \quad \forall\, n,\, k.$$

6.4 Vectorized formulation

Using $\mathrm{vec}\big(\mathcal{S} \times_1 D^{(1)} \cdots \times_N D^{(N)}\big) = \big(D^{(N)} \otimes \cdots \otimes D^{(1)}\big)\, \mathrm{vec}(\mathcal{S})$:

$$\mathrm{vec}(\mathcal{Y}_i) \approx \underbrace{\big(D^{(N)} \otimes \cdots \otimes D^{(1)}\big)}_{\mathbf{D} \in R^{(\prod I_n) \times (\prod J_n)}}\, \mathrm{vec}(\mathcal{S}_i). \tag{13}$$

The Kronecker-structured dictionary $\mathbf{D}$ is **never formed explicitly**.

7 Adaptive-Rank TT-DL (Project 2)

Adaptive-Rank TT-DL Optimization (Project 2):

$$\min_{D^{(1:N)}\{C_i^{(1:N)}\}} \sum_{i=1}^{m} \frac{1}{2} \big\| \mathcal{Y}_i - \mathrm{Recon}(\cdot) \big\|_{\mathrm{F}}^2 + \lambda_1 \sum_{i,n} \big\| C_i^{(n)} \big\|_1 + \lambda_2 \sum_{n=1}^{N-1} \Phi(\rho_n) \tag{14}$$

$$\text{subject to} \quad \|d_k^{(n)}\|_2 \le 1 \quad \forall\, n, k.$$

Three choices for the rank penalty Φ:

1. **Nuclear norm:** $\Phi(\rho_n) = \sum_{i=1}^{m} \big\|C_{i\,(1)}^{(n)}\big\|_*$, where $\|\cdot\|_*$ is the nuclear norm of the left unfolding.
2. **Log-determinant:** $\Phi(\rho_n) = \sum_{i=1}^{m} \log\det\big(C_{i\,(1)}^{(n)\mathsf{T}} C_{i\,(1)}^{(n)} + \varepsilon I\big)$.
3. **Explicit SVD truncation** (non-differentiable; see Section 11).

8 Alternating Minimization Algorithm

Algorithm: TT-DL via Alternating Minimization with Adaptive Ranks
Require: Data $\{\mathcal{Y}_i\}_{i=1}^{m}$; sizes $J_1, \ldots, J_N$; initial ranks ρ; params λ_1, λ_2, T

1. Init $D^{(n)}$ with random orthonormal columns; init $\{C_i^{(n)}\}$ at ranks ρ
2. **For** $t = 1, \ldots, T$:

   Phase 1: Sparse Coding (ALS Sweep)

   - For $i = 1, \ldots, m$: For $n = 1 \to N$ then $N \to 1$: Compute $L_i^{(n)}$, $R_i^{(n)}$; solve local LASSO eq:local-lasso; QR-orthogonalize

   Phase 2: Dictionary Update

   - For $n = 1, \ldots, N$: Compute $\tilde{Z}^{(n)}$; update $D^{(n)}$ via eq:dict-pgd

   Phase 3: Rank Adaptation

   - For $n = 1, \ldots, N-1$: Merge, SVD, threshold, truncate/enrich ρ_n

   If converged, break.

9 Sparse Coding Sub-problem (Phase 1)

9.1 Partial contractions

Fix all TT-cores except $C_i^{(n)}$. The **left partial contraction** for modes $1, \ldots, n-1$:

$$L_i^{(n)}(i_1, \ldots, i_{n-1};\ \alpha_{n-1}) = \sum_{j_1,\ldots,j_{n-1}\,\alpha_1,\ldots,\alpha_{n-2}} \prod_{k=1}^{n-1} \Big[C_i^{(k)}(\alpha_{k-1}, j_k, \alpha_k)\ D^{(k)}(i_k, j_k)\Big], \tag{15}$$

yielding $L_i^{(n)} \in R^{(\prod_{k<n} I_k)\times\rho_{n-1}}$. The right contraction $R_i^{(n)} \in R^{\rho_n\times(\prod_{k>n} I_k)}$ is defined analogously.

9.2 Reduction to LASSO

Key Insight: The reconstruction is *linear* in $\mathrm{vec}(C_i^{(n)})$:

$$\mathrm{vec}(\mathcal{Y}_i) \approx \underbrace{\big(R_i^{(n)} \otimes D^{(n)} \otimes L_i^{(n)}\big)}_{A_i} \mathrm{vec}\big(C_i^{(n)}\big). \tag{16}$$

Local LASSO at Mode n:

$$\min_{\breve{c}_i} \frac{1}{2}\big\|\breve{y}_i - A_i\,\breve{c}_i\big\|_2^2 + \lambda_1 \|\breve{c}_i\|_1, \tag{17}$$

where $\breve{y}_i = \mathrm{vec}(\mathcal{Y}_i)$ and $\breve{c}_i = \mathrm{vec}(C_i^{(n)})$.

ISTA solver.

$$\breve{c}_i \leftarrow \mathrm{soft}_{\lambda_1/L_i}\Big(\breve{c}_i + 1L_i\, A_i^{\mathsf{T}}\big(\breve{y}_i - A_i\,\breve{c}_i\big)\Big), \tag{18}$$

where $L_i = \|A_i\|^2$ and $\mathrm{soft}_\tau(x) = \mathrm{sign}(x) \cdot \max(|x| - \tau,\, 0)$.

Efficient computation. The Kronecker-structured A_i is **never formed**. Products $A_i\breve{c}_i$ cost $\mathcal{O}\big(\rho^2 J \cdot \max(\prod_{k<n} I_k,\ \prod_{k>n} I_k)\big)$.

9.3 ALS sweep and orthogonalization

After optimizing $C_i^{(n)}$ in a left-to-right sweep:

$$C_{i\,(1)}^{(n)} = QR, \qquad C_i^{(n)} \leftarrow \mathrm{reshape}(Q), \qquad C_i^{(n+1)} \leftarrow R \cdot C_i^{(n+1)}. \tag{19}$$

Remark. This ALS sweep with orthogonalization is directly analogous to the **DMRG** algorithm from quantum many-body physics.

10 Dictionary Update Sub-problem (Phase 2)

Fix all TT-cores and all dictionaries except $D^{(n)}$. Stacking mode-n unfoldings across samples:

$$\tilde{Y}^{(n)} \approx D^{(n)}\,\tilde{Z}^{(n)}. \tag{20}$$

Dictionary Update at Mode n:

$$\min_{D^{(n)}} \frac{1}{2}\big\|\tilde{Y}^{(n)} - D^{(n)}\tilde{Z}^{(n)}\big\|_{\mathrm{F}}^2 \qquad s.t. \quad \|d_k^{(n)}\|_2 \le 1. \tag{21}$$

Projected gradient descent.

$$D^{(n)} \leftarrow D^{(n)} - \eta\,\big(D^{(n)}\tilde{Z}^{(n)} - \tilde{Y}^{(n)}\big)\,\tilde{Z}^{(n)\mathsf{T}}, d_k^{(n)} \leftarrow d_k^{(n)} \Big/ \max\big(1,\ \|d_k^{(n)}\|_2\big). \tag{22}$$

11 Adaptive Rank Selection (Phase 3)

For each bond $n = 1, \ldots, N-1$:

Step 1: Merge. $M_i^{(n)} = \sum_{\alpha_n} C_i^{(n)}(\cdot,\cdot,\alpha_n)\ C_i^{(n+1)}(\alpha_n,\cdot,\cdot)$. Reshape to $\tilde{M}_i^{(n)} \in R^{(\rho_{n-1}J_n)\times(J_{n+1}\rho_{n+1})}$.

Step 2: SVD. $\tilde{M}_i^{(n)} = U_i\,\Sigma_i\,V_i^{\mathsf{T}}$.

Step 3: Threshold.

1. Relative: $\rho_n^{\text{new}} = \#\{k : \sigma_k/\sigma_1 > \delta\}$.
2. Energy: $\rho_n^{\text{new}} = \min\{r : \sum_{k=1}^{r}\sigma_k^2/\sum_k \sigma_k^2 > 1-\varepsilon\}$.
3. BIC: $\rho_n^{\text{new}} = \arg\min_r \big[m\|\text{res}(r)\|_{\text{F}}^2 + r\log m\big]$.

Step 4: Truncate/enrich. Keep top ρ_n^{new} singular triplets, or enrich by projecting the residual onto the orthogonal complement of the current TT manifold.

12 Computational Complexity

Complexity Comparison:

	Tucker DL	**TT-DL (ours)**
Sparse coding	$\mathcal{O}(m\cdot J^N\cdot K_{ISTA})$	$\mathcal{O}(m\,N\,\rho^2 J\,K_{ISTA})$
Dictionary update	$\mathcal{O}(N\,m\,I^N J^N)$	$\mathcal{O}(N\,m\,\rho^2 IJ)$
Storage	$m\cdot J^N$	$m\cdot N\cdot\rho^2\cdot J$
Scaling in N	**Exponential**	**Linear**

Proposition (Speedup). $Tuckercost\,/\,TT-DLcost \approx J^N/(N\rho^2 J)$.

Example. N=6, J=15, ρ=5: speedup $\approx$ 5,000×. N=10, J=10, ρ=5: speedup $\approx 4\times10^6$.

13 Extension: Deep Unrolled TT-DL

Unroll T iterations into a T-layer network with learnable ISTA at each layer:

$$\breve{c}_i^{(t)} \leftarrow \text{soft}_{\theta^{(t)}}\Big(\breve{c}_i^{(t-1)} + W_s^{(t)}\,A_i^{\mathsf{T}}\big(\breve{y}_i - A_i\,\breve{c}_i^{(t-1)}\big)\Big), \tag{23}$$

where $W_s^{(t)}$ (step-size) and $\theta^{(t)}$ (threshold) are *learnable per layer*. Dictionary updates use learnable step sizes $\eta^{(t)}$. Loss:

$$\mathcal{L} = \sum_{i=1}^{m} \left\| \mathcal{Y}_i - \mathrm{Recon}\big(D^{(T)}, C_i^{(T)}\big) \right\|_{\mathrm{F}}^2. \tag{24}$$

Classical TT-DL needs T=100+ iterations; the unrolled version converges in T=5–15 layers.

14 Research Roadmap

#	Project	Novelty	Key Result
1	TT-DL, high-order data	Eq. eq:ttdl-opt	Tucker fails at order $\geq$5; TT-DL works at 10+
2	Adaptive-rank TT-DL	Section 11	Auto rank selection improves efficiency & accuracy
3	Topology-adaptive TN-DL	TT/HT/TR comparison	Data-dependent tensor network structure
4	Deep unrolled TT-DL	Eq. eq:unrolled-ista	5 layers match 100-iter classical TT-DL
5	Sample complexity	Minimax bounds	$\mathcal{O}(NJr^2)$ sample complexity for TT-DL

Recommended path: Projects 1+2 form a strong first paper. Project 4 targets the deep learning community. Project 5 provides theoretical foundations.

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